

Research Interests

My long-term research interest lies in **measure-theoretic questions about language**, especially how measures over language shape model intelligence, generalization, and training efficiency in reinforcement learning. My work on **non-autoregressive modeling** and **post-training** follows this same line of inquiry.

I develop practical models and algorithms grounded in theoretical insight, with applications to reinforcement learning, continual and test-time learning, controllable generation, and language-model architectures beyond autoregressive Transformers.

Research and Professional Experience

Seed LLM

RESEARCH LEAD FOR REAL-TIME CREATIVE MUSIC GENERATION

- Leading research on real-time creative music generation.

San Jose, CA, U.S.A.

Apr. 2026 – Present

Tencent AI Lab Seattle

FULL-TIME RESEARCH SCIENTIST

- Developed Locas and G2RL in close collaboration with Zhenwen Liang.
- Mentored Mohan Tang on Turbo Connection.

Bellevue, WA, U.S.A.

Aug. 2025 – Mar. 2026

careerdog.ai (Startup)

TECH LEAD (PRO BONO)

- Designed a Browser-Use Web Agent integrating job search, application form filling, and automated drafting of application materials using early-generation LLMs.
- Achieved over 10× the speed of competing products at the time through system design.

Remote

2023

Amazon Devices / AGI

RESEARCH INTERN

- Studied applications and practical boundaries of controllable-generation methods.

Sunnyvale, CA, U.S.A.

Summer 2021, 2022, and 2023

PlusLab, UCLA / USC

PH.D. RESEARCH ASSISTANT

- Conducted research under the supervision of Professor Nanyun Peng.
- Studied machine-learning foundations for natural-language generation, including non-autoregressive models and controllable generation.

Los Angeles, CA, U.S.A.

Jun. 2019 – Jun. 2025

CSAIL, Massachusetts Institute of Technology

VISITING STUDENT

- Collaborated with members of Professor Joshua B. Tenenbaum's research group on combinatorial generalization and Bayesian program learning.

Cambridge, MA, U.S.A.

Jul. 2018 – Jan. 2019

APEX Lab, Shanghai Jiao Tong University

UNDERGRADUATE RESEARCHER

- Worked with Professors Weinan Zhang and Yong Yu on discrete text generation and reinforcement learning.

Shanghai, P.R.C.

Jun. 2017 – Jun. 2019

Education

University of California, Los Angeles

PH.D. IN COMPUTER SCIENCE (NATURAL LANGUAGE PROCESSING; MINOR IN THEORY)

- Advisor: Professor Nanyun Peng.
- Thesis: *Towards More Flexible and Controllable Language Models*.

Los Angeles, CA, U.S.A.

Sept. 2020 – Jun. 2025

University of Southern California

PH.D. STUDENT IN COMPUTER SCIENCE

- Transferred to UCLA with advisor Nanyun Peng.

Los Angeles, CA, U.S.A.

Sept. 2019 – Jul. 2020

ACM Honors Class, Zhiyuan College, Shanghai Jiao Tong University

B.S. IN COMPUTER SCIENCE (ZHIYUAN HONORS PROGRAM)

- Selected for the ACM Honors Class.

Shanghai, P.R.C.

Sept. 2015 – Jul. 2019

Selected Publications

Turbo Connection: Reasoning as Information Flow from Higher to Lower Layers

Mentor / Preprint

MOHAN TANG, **SIDI LU**

2026

- Mentored the first author and guided the project's research direction.
- Introduces cross-token, higher-to-lower residual routing that lengthens the effective computation path for multi-step reasoning.

Locas: Your Models are Principled Initializers of Locally-Supported Parametric Memories

First Author / Tencent AI Lab

Technical Report

SIDI LU, ZHENWEN LIANG, DONGYANG MA, YAN WANG, HAITAO MI, DONG YU

2026

- Developed locally-supported parametric memories that can be attached for continual learning and later offloaded or merged into a model's feed-forward parameters.
- Locas-GLU retains past context with 0.02% additional parameters while preserving general capability.

Can LLMs Guide Their Own Exploration? Gradient-Guided Reinforcement Learning for LLM Reasoning

Second Author / Preprint

ZHENWEN LIANG, **SIDI LU**, WENHAO YU, KISHAN PANAGANTI, YUJUN ZHOU, HAITAO MI, DONG YU

2025

- Developed a reinforcement-learning framework that rewards trajectories contributing useful new gradient directions.
- Contributed the gradient-centric exploration idea, theoretical analysis, and algorithm design.

InsNeXt: Training Scalable Insertion-based Language Models from Scratch

First Author / COLM 2026 Workshop

SIDI LU, JACKY DAI, XUEZHE MA, NANYUN PENG

2026

- Built a scalable insertion-based pretraining pipeline reaching 573M parameters and a 4,096-token context window.

Save the Good Prefix: Precise Error Penalization via Process-Supervised RL to Enhance LLM Reasoning

Findings of ACL 2026

HAOLIN LIU, DIAN YU, **SIDI LU**, YUJUN ZHOU, RUI LIU, ZHENWEN LIANG, HAITAO MI, CHEN-YU WEI, DONG YU

2026

Too Correct to Learn: Reinforcement Learning on Saturated Reasoning Data

ACL 2026 Short Papers

ZHENWEN LIANG, YUJUN ZHOU, **SIDI LU**, XIANGLIANG ZHANG, HAITAO MI, DONG YU

2026

DiNADO: Norm-Disentangled Neurally-Decomposed Oracles for Controlling Language Models

First Author / ICML 2024

SIDI LU, WENBO ZHAO, CHENYANG TAO, ARPIT GUPTA, SHANCHAN WU, TAGYOUNG CHUNG, NANYUN PENG

2024

Open-Domain Text Evaluation via Contrastive Distribution Methods

First Author / ICML 2024

SIDI LU, HONGYI LIU, ASLI CELIKYILMAZ, TIANLU WANG, NANYUN PENG

2024

Controllable Text Generation with Neurally-Decomposed Oracle

Co-First Author / NeurIPS 2022

Highlighted Paper

TAO MENG*, **SIDI LU***, NANYUN PENG, KAI-WEI CHANG

2022

InsNet: An Efficient, Flexible, and Performant Insertion-based Text Generation Model

First Author / NeurIPS 2022

SIDI LU, TAO MENG, NANYUN PENG

2022

CoT: Cooperative Training for Generative Modeling of Discrete Data

First Author / ICML 2019 Short Oral

SIDI LU, LANTAO YU, SIYUAN FENG, YAOMING ZHU, WEINAN ZHANG

2019

Neurally-Guided Structural Inference

Sidi Lu*, Jiayuan Mao*, Joshua B. Tenenbaum, Jiajun Wu

Co-First Author / ICML 2019 Short
Oral
2019

Long Text Generation via Adversarial Training with Leaked Information

Jiaxian Guo, Sidi Lu, Han Cai, Weinan Zhang, Yong Yu, Jun Wang

Second Author / AAAI 2018
2018

Texygen: A Benchmarking Platform for Text Generation Models

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, Yong Yu

Second Author / SIGIR 2018 Short
Paper
2018

Neural Text Generation: Past, Present and Beyond

Sidi Lu, Yaoming Zhu, Weinan Zhang, Jun Wang, Yong Yu

Preprint
2018

Patent

Gradient Diversity in Reinforcement Learning of Large Language Models

Zhenwen Liang, Sidi Lu

- Contributed to the theoretical analysis, algorithm design, and prototype development.

U.S. Patent Application
Submitted Nov. 2025

Teaching Experience

TEACHING ASSISTANT

2016	Junior TA, Programming
2017	Senior TA, Data Structure
2017	Senior TA, Programming Practice
2018	Senior TA, Computer System (II): Operating System
2018	Junior TA, Compiler: Practice and Design
2020	TA, CS269 Computer Science Seminar: Natural Language Generation
2024	TA, CS96 Introduction to Generative AI

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